

46 YEARS OF EXCELLENCE | SRAMAYEVA JAYATHE | NARAYANA IIT - JEE ACADEMY - INDIA

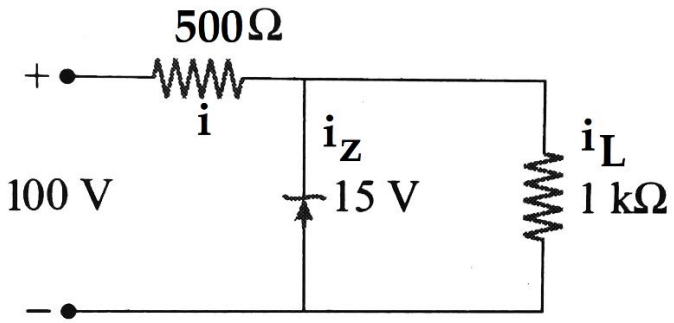
Sec: SR.IIT_*CO-SC(MODEL-A& APEX) **SPCL_GTM-05 (FN)** **Date: 15.01.2025**
Time: 3.00Hrs. **JEE MAIN MODEL** **Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

01. A Zener diode, having breakdown voltage equal to 15V, is used in a voltage regulator circuit shown in figure. The current through the diode is



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A) 5 mA B) 150 mA C) 15 mA D) 155 mA

02. The electric current through a wire varies with time as $I = I_0 - \beta t$. Where $I_0 = 20\,A$ and $\beta = 4A/s$. The amount of electric charge crossed through a section of the wire in 5 sec is:

A) 50 C B) 150 C C) 500 C D) 1500 C

03. **Statement 1:** A uniform elastic rod lying on smooth horizontal surface is pulled by constant horizontal force of magnitude F as shown in figure (i). Another identical elastic rod is pulled vertically upwards by a constant vertical force of magnitude F (figure ii). The extension in both rods will be same.

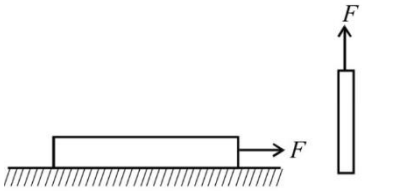


Fig. (i) Fig. (ii)

Statement-2: In above cases, the extension depends only on constant force F acting at the end of the rods.

A) Both Statement I and Statement II are true

B) Both Statement I and Statement II are false

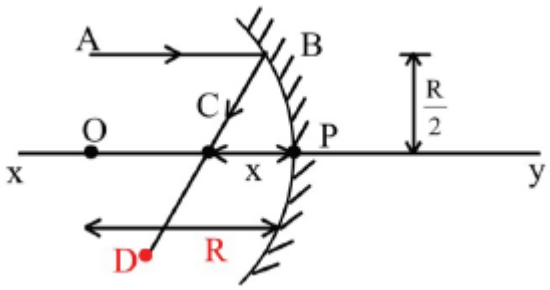
C) Statement I is true but Statement II is false

D) Statement I is false but Statement II is true

04. A light ray (AB) incident parallel to principal axis (XY) at a distance of R/2 (R is radius of curvature of the concave mirror) from principal axis as shown. BCD is the reflected ray which intersects principal axis at C. The value of x, if O be centre of curvature and P is the pole of the mirror, is

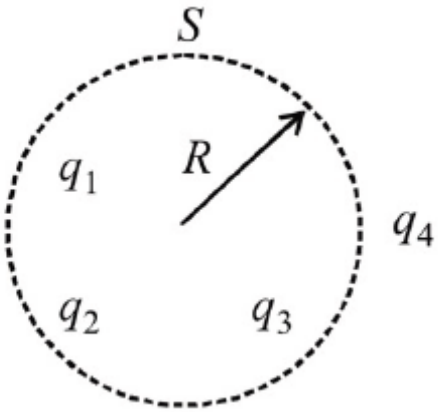
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A) $\frac{R}{2}$ B) $\frac{3R}{8}$ C) $R\left(1-\frac{1}{\sqrt{3}}\right)$ D) $\frac{R}{\sqrt{3}}$

05. q_1, q_2, q_3 and q_4 are point charges located as shown in the figure and S is a spherical Gaussian surface of radius R. Which of the following is true according to the Gauss's law:



A) $\oint\left(\vec{E}_1+\vec{E}_2+\vec{E}_3\right).d\vec{A}=\frac{q_1+q_2+q_3}{\in_0}$

B) $\oint\left(\vec{E}_1+\vec{E}_2+\vec{E}_3+\vec{E}_4\right).d\vec{A}=\frac{q_1+q_2+q_3}{\in_0}$

C) $\oint\left(\vec{E}_1+\vec{E}_2+\vec{E}_3+\vec{E}_4\right).d\vec{A}=\frac{q_1+q_2+q_3+q_4}{\in_0}$

D) $\oint\left(\vec{E}_1+\vec{E}_2+\vec{E}_3+\vec{E}_4\right).d\vec{A}=\frac{q_1-q_2-q_3+q_4}{\in_0}$

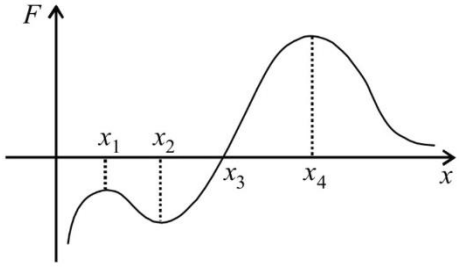
06. A point moves in x-y plane according to the law $x=3\cos 4t$ and $y=3(1-\sin 4t)$. The distance travelled by the particle in 2 sec is (where x and y are in meters)

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A) 48 m B) 24 m C) $48\sqrt{2}\,m$ D) $24\sqrt{2}\,m$

07. The adjacent figure shows an interaction force between two particles A and B against the separation between them, when the separation between them changes from zero to infinity. Parts of the curve above and below the x-axis denote attractive and repulsive forces, respectively. Then which of the following statement is incorrect?



A) Kinetic energy of the system increases from $x_1 \, to \, x_2$

B) Potential energy of the system decreases from $x_1 \, to \, x_2$

C) Kinetic energy of the system increases from $x_3 \, to \, x_4$

D) Kinetic energy of the system decreases from $x_3 \, to \, x_4$

08. In the experiment of Ohm's law, when potential difference of 10.0 V is applied, current measured is 1.00A. If the length of wire is found to be 10.0 cm and diameter of wire 2.50 mm, then the maximum permissible percentage error in resistivity is;

A) 3.1% B) 3.8% C) 3.2% D) 3.01%

09. Two blocks of masses $m_1 = 10\,kg$ and $m_2 = 20kg$ are connected by a spring of stiffness $k = 200\,\text{N/m}$. The coefficient of friction between the blocks and the fixed horizontal surface $\mu = 0.1$. Find the minimum constant horizontal force is F (in Newtons) to be applied on m_1 in order to slide the mass m_2. [Take $g = 10\,m/s^2$]

A) 20 B)15 C) 7 D) 6

10. A light beam travelling in the x-direction is described by the electric field $E_y = (300V/m)\sin\omega\left(t - \frac{x}{C}\right)$. An electron is constrained to move along the y-direction with a speed of $2.0\times10^7\,m/s$.

APEX JEE-MAIN 2025

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The maximum magnetic force on the electron is___ $\times 10^{-18} N$. ($C = 3 \times 10^8 m/s$)

($e = 1.6 \times 10^{-19} C$)

A) 2.5 B) 2.0 C) 4.0 D) 3.2

11. Body moves in a circular path of radius R with deceleration so that at any moment of time its tangential and normal accelerations are equal in magnitude. At the initial moment t=0, the velocity of body is v_0, then the velocity of body as a function of distance travelled S will be;

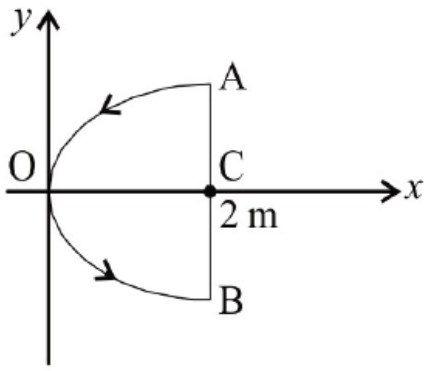
A) $\upsilon = \upsilon_0 e^{S/R}$ after it has moved S meter.

B) $\upsilon = \upsilon_0 e^{\frac{-S}{R}}$ after it has moved S meter.

C) $\upsilon = \upsilon_0 e^{-SR}$ after it has moved S meter.

D) $\upsilon = \upsilon_0 e^{\frac{-R}{S}}$ after it has moved S meter.

12. A conducting wire is bend into a loop as shown in the figure. The segments AOB is parabolic given by the equation $y^2 = 2x$ while segment BA is a straight line parallel to the y-axis. The magnetic field in the region is $\vec{B} = -8\hat{k}$ T and the current in the wire is 2A. Then The torque on the loop will be;



A) $16\sqrt{2} Nm$ B) $16 Nm$ C) $18\sqrt{2} Nm$ D) Zero

13. A galvanometer of resistance 5 ohms gives a full-scale deflection for a potential difference of 10 mV. To convert the galvanometer into a voltmeter giving a full-scale deflection for a potential difference of 1 V, the resistance that must be attached to the

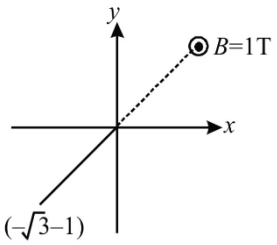
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galvanometer in series is;

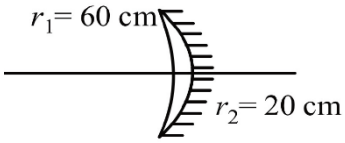
A) 0.495 ohm B) 49.5 ohm C) 495 ohm D) 4950 ohm

14. A uniform magnetic field of magnitude 1T which exists in region y≥0 is along $\hat{k}$ direction as shown. A particle of charge 1C is projected from point $\left(-\sqrt{3},-1\right)$ towards origin with speed 1 m/sec. If mass of particle is 1 kg, then co-ordinates of centre of circle in which particle moves are



A) $\left(-\frac{1}{2},-\frac{\sqrt{3}}{2}\right)$ B) $\left(-\frac{\sqrt{3}}{2},\frac{1}{2}\right)$ C) $\left(\frac{\sqrt{3}}{2},-\frac{1}{2}\right)$ D) $\left(\frac{1}{2},-\frac{\sqrt{3}}{2}\right)$

15. Convex surface of thin concavo convex lens of refractive index 1.5 is silvered as shown. A small object is kept in air at 30 cm left of the lens on its principal axis. The distance of the final image is



A) 20 cm B) 30 cm C) 10 cm D) 15 cm

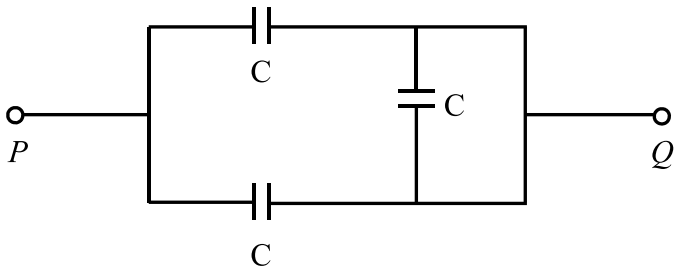
16. A bullet is fired from a gun. The force on the bullet is given by $F=600-\left(2\times10^{5}\right)t$ Here, F is in newton and t in second. The force on the bullet becomes zero as soon as it leaves th barrel. The impulse imparted to the bullet is

A) $8N-s$ B) $0.9N-s$ C) $1.8N-s$ D) $2.4N-s$

APEX JEE-MAIN 2025

For More Material Join: @JEEAdvanced_2025

17. The equivalent capacitance between points P and Q of the combination shown in figure is



A) C B) $2C$ C) $\frac{3}{2}C$ D) $\frac{C}{2}$

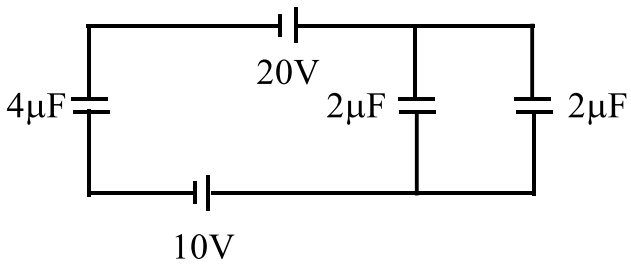
18. The maximum attainable temperature of an ideal gas in the process $P = P_0 - \alpha V^2$, where P_0 and α are +ve constant is:

A) $\frac{2P_0}{3nR}\left(\frac{P_0}{3\alpha}\right)^{1/2}$ B) $\frac{P_0}{2nR}\left(\frac{2P_0}{3\alpha}\right)^{1/2}$ C) $\frac{2nR}{P_0}\left(\frac{2P_0}{3\alpha}\right)^{1/2}$ D) $\frac{2P_0}{nR}\left(\frac{P_0}{2\alpha}\right)^{1/2}$

19. If g is the acceleration due to gravity on the earth's surface, the change in the potential energy of an object of mass m raised from the surface of the earth to a height equal to the radius R of the earth is

A) $mgR/2$ B) $2mgR$ C) mgR D) $-mgR$

20. In the circuit shown in figure charge stored in $4\ \mu F$ capacitor is



A) $20\ \mu F$ B) $40\ \mu F$ C) $10\ \mu F$ D120 μF

APEX JEE-MAIN 2025

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SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only.** **Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

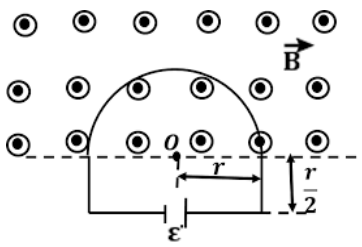
21. The binding energy of deuteron $({}_1^2H)$ is 1.15 MeV per nucleon and an alpha particle $({}_2^4He)$ has a binding energy of 7.1 MeV per nucleon. Then in the given reaction the energy Q released is x /10 MeV. Then find the value of x.

$${}_1^2H+{}_1^2H\longrightarrow{}_2^4He+Q$$

22. When a metallic surface is illuminated with light of wavelength λ, the stopping potential is V. When the same surface is illuminated by light of wavelength 2λ, the stopping potential is $\frac{V}{3}$. The threshold wavelength for the surface is $x\lambda$, then the value of x is

23. A body is projected horizontally from the top of a tower with initial velocity 18m/s. It hits the ground at an angle 45°. What is the magnitude of vertical component of velocity (in m/s) when it strikes the ground?

24. The figure shows a conducting loop consisting of half circle of area $A=0.06\,m^2$ and three straight segments. The half circle lies in a uniform changing magnetic field $B=4t^2+2t+5$(SI unit), where t is the time in second. An ideal battery of emf E= 2 V is connected as shown and the total resistance of the wire is 2Ω. The net current in the loop at t=5 second is x /100 A. Then find the value of x.



APEX JEE-MAIN 2025

For More Material Join: @JEEAdvanced_2025

25. A body is executing simple harmonic motion. At a displacement 'x' from mean position its potential energy is $E_1 = 2J$ and at a displacement 'y' from mean position, its potential energy is $E_2 = 8J$ The potential energy 'E' (in J) at a displacement $(x+y)$ from mean position is.

APEX JEE-MAIN 2025

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CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. What is the correct increasing order of extent of deprotonation of the following acids in their aqueous solution?

I. $(CH_3)_2CHCOOH$ II. CH_3CH_2COOH

III. Cl_2CHCH_2COOH IV. CH_3CCl_2COOH

A) $I < II < III < IV$ B) $IV < III < I < II$

C) $II < I < III < IV$ D) $I < IV < III < II$

27. Which of the following statements is not correct?

A) Number of H-bonds between Adenine and Thymine are 2.

B) $\alpha - D$ Glucopyranose and $\beta - D$ Glucopyranose are anomers.

C) Sucrose is non reducing sugar

D) All amino acids except Glycine are optically inactive

28. Match the following:

COLUMN-I (Compounds)		**COLUMN-II (Differentiating Reagent/test)**	
I	1-Butyne & 2- Butyne	P	Gives off CO_2 with $NaHCO_3$
II	Ethyl alcohol & n-Propyl alcohol	Q	Ammonical Cu_2Cl_2
III	Glucose & Fructose	R	NaOI
IV	Picric acid & o-nitro phenol	S	Seliwanoff test

A)I – P; II- R; III- S; IV – Q B)I – R; II- Q; III- S; IV - P

C)I – Q; II- R; III- S; IV – P D)I – S; II- R; III- Q; IV – P

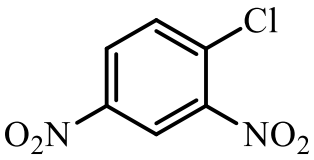
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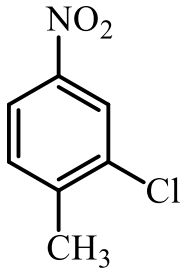
29. If the concentration of CrO_4^{-2} ions in a saturated solution of Ag_2CrO_4 is 2×10^{-4} mol. lit^{-1}. The K_{Sp} of Ag_2CrO_4 will be

A) 1.6×10^{-8} B) 3.2×10^{-11} C) 4.6×10^{-6} D) 4.8×10^{-12}

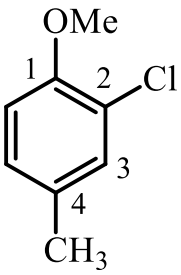
30. Among the following which is the correct IUPAC name of the following?



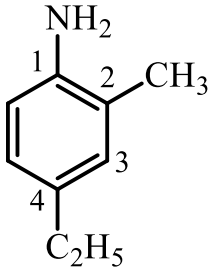
(i) 1-Chloro-2,4-dinitrobenzene



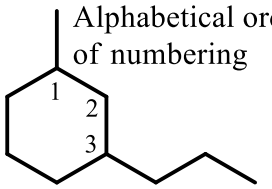
(ii) 2-Chloro-1-methyl-4-nitrobenzene



(iii) 2-Chloro-4-methylanisole



(iv) 4-Ethyl-2-methylaniline



(v) 1-Methyl-3-propylcyclohexane

A) (i) and (iii) only B) (i) and (v) only

C) (i), (ii), (iii), (iv) and (v) D) (iii) and (v) only

31. What is the IUPAC name of compound $NaBH_4$?

A) Sodium boronhydride B) Sodium tetrahydridoboron (II)

C) Sodium tetrahydridoborate (III) D) Sodium tetrahydridoborate (I)

APEX JEE-MAIN 2025

For More Material Join: @JEEAdvanced_2025

32. An aqueous solution is a mixture of acetates of Zinc, Nickel and Copper. In all acetates oxidation states of metal is $+2$. If H_2S gas is passed through the solution, the ions that get precipitated as sulphide are

A) Zn^{2+}, Ni^{2+} B) Cu^{2+}, Ni^{2+} C) Zn^{2+}, Cu^{2+} D) Zn^{2+}, Cu^{2+} and Ni^{2+}

33 Match the following.

	Column – I		**Column – II**
P.	$C_6H_5CONH_2 \xrightarrow[\Delta]{P_2O_5}$	I.	C_6H_5CHO
Q.	$C_6H_5COCl \xrightarrow[Pd-BdSO_4]{H_2}$	II.	$C_6H_5CH_2NH_2$
R.	$CH_3COOCH_2C_6H_5 \xrightarrow[H_3O^+]{LiAlH_4}$	III.	C_6H_5CN
S.	$C_6H_5CONH_2 \xrightarrow[H_3O^+]{LiAlH_4}$	IV.	$C_6H_5CH_2OH$

A) $P(III), Q(I), R(IV), S(II)$ B) $P(II), Q(I), R(IV), S(III)$

C) $P(II), Q(IV), R(III), S(I)$ D) $P(III), Q(II), R(I), S(IV)$

34. Match the equation in Column I with the name type in Column II.

COLUMN-I		**COLUMN-II**	
I.	$\Delta x \Delta v \geq \frac{h}{4\pi m}$	P.	Rydberg equation
II.	$mvr = \frac{nh}{2\pi}$	Q.	De-Broglie equation
III.	$\lambda = \frac{h}{\sqrt{2mKE}}$	R.	Angular momentum is quantised
IV.	$\bar{v} = R\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$	S.	Heisenberg's uncertainty principle

A) $I-S, II-R, III-Q, IV-P$ B) $I-S, II-Q, III-R, IV-P$

C) $I-Q, II-P, III-S, IV-R$ D) $I-S, II-P, III-Q, IV-R$

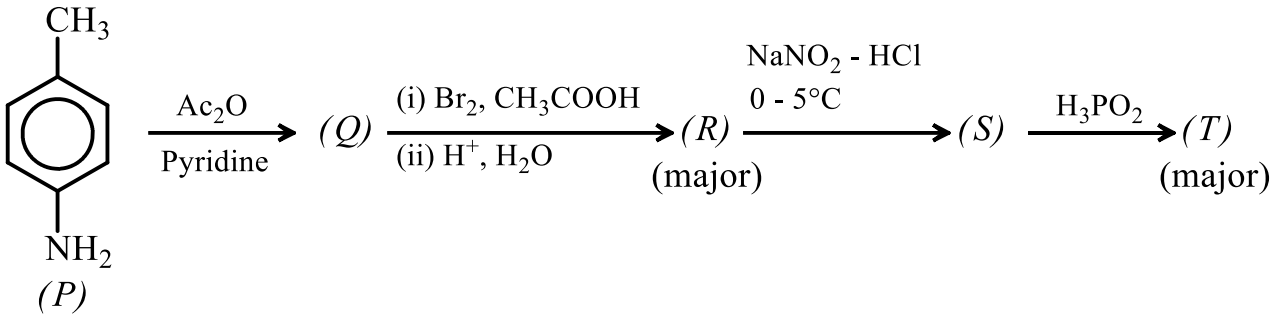
APEX JEE-MAIN 2025

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35. The conductivity of a saturated solution of $BaSO_4$ is $3.06\times10^{-6} Ohm^{-1}cm^{-1}$ and its equivalent conductance is $1.53\, Ohm^{-1}cm^2\ gr\ eq^{-1}$. K_{sp} of $BaSO_4$ is___

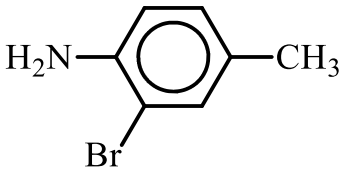
A) 2×10^{-3} B) 10^{-3} C) 10^{-6} D) 2×10^{-6}

36. Consider the given reaction sequence

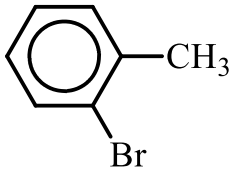


Identify the incorrect statement.

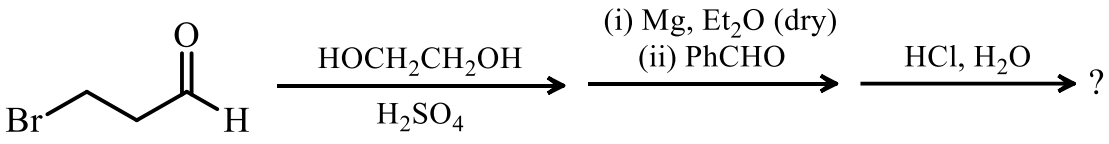
A) (P) can be named as p-Toluidine

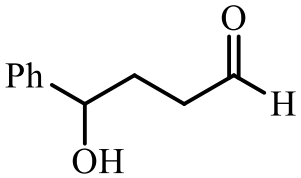
B) (R) is 

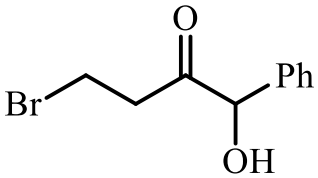
C) The step (R) to (T) carries out deamination

D) (S) is 

37. What is the major organic product obtained from the following sequence of reaction?

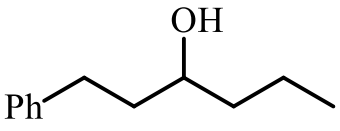


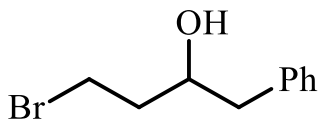
I) 

II) 

APEX JEE-MAIN 2025

For More Material Join: @JEEAdvanced_2025

III) 

IV) 

A) Only I B) Only II C) Only III D) Only IV

38. Which of the following statement is false?

A) With fluorine vanadium can form VF_5

B) With chlorine vanadium can form VCl_5

C) Vanadium exhibits highest oxidation state in oxohalides $VOCl_3$, $VOBr_3$ and fluoride VF_5

D) With iodine vanadium cannot form VI_5 due to oxidation power of V^{5+} and reducing nature of I^-

39. Which of the following acts as the stationary phase in paper chromatography?

A) Water B) Alumina C) Silica gel D) None of these

40. Match the following characteristics given in Column I with the corresponding elements given in Column II. Mark the correct option from the codes given below.

	Column I		**Column II**
i.	Element with low enthalpy of atomization	p.	*Cu*
ii.	Element with highest second ionization energy	q.	*Zn*
iii.	Element with high melting point	r.	*Sc*
iv.	Element with highest metallic radius	s.	*Cr*

A) (i) → (s), (ii) → (p), (iii) → (r), (iv) → (q)

B) (i) → (q), (ii) → (p), (iii) → (r), (iv) → (s)

C) (i) → (p), (ii) → (r), (iii) → (q), (iv) → (s)

D) (i) → (q), (ii) → (p), (iii) → (s), (iv) → (r)

APEX JEE-MAIN 2025

For More Material Join: @JEEAdvanced_2025

41. I) $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$; Equilibrium constant $= K_1$

II) $H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$; Equilibrium constant $= K_2$

III) $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$; Equilibrium constant $= K_3$

Assuming equilibrium concentration of above every species is $0.01\,M$ and arrange these reactions in the increasing order of extent of completion of reaction

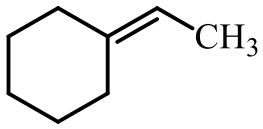
A) $I < II < III$ B) $III < II < I$ C) $I < III < II$ D) $II < I < III$

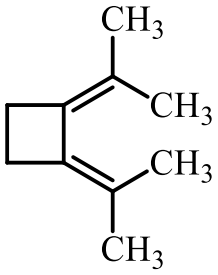
42. Pick out the compound which reacts fastest in the presence of $AgNO_3$.

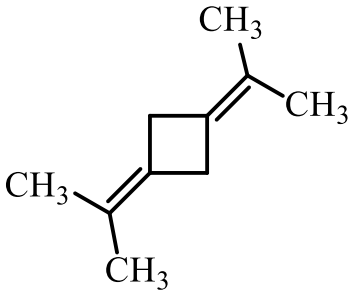
A) $(CH_3)_3CCl$ B) $(CH_3)_2CHCH_2Cl$

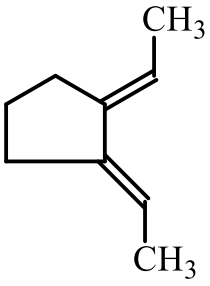
C) $(CH_3)_2CHCl$ D) CH_3CH_2Cl

43. Which compound below can show geometrical isomerism?

A) 

B) 

C) 

D) 

44. **Statement I:** Fluorine has less negative electron gain enthalpy than Chlorine

Statement II: Additional electron is repelled more effectively by $3p$ electrons in Chlorine than by $2p$ electrons in Fluorine.

A) Both Statement I and Statement II are correct and Statement II is the correct explanation of Statement I

B) Both Statement I and Statement II are correct but Statement II is not the correct

APEX JEE-MAIN 2025

For More Material Join: @JEEAdvanced_2025

explanation of Statement I

C) Statement I is correct but Statement II is incorrect

D) Statement II is correct but Statement I is incorrect

45. The number of non-chiral and chiral isomers for the complex $\left[Ma_2b_2c_2\right]^{\pm n}$ in which a,b,c are monodentate ligands, are

A) 4,2 B) 6,2 C) 6,4 D) 6,3

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

46. How many bonding MO are used in the formation of NO?

47. At 298 K , the equilibrium constant is 2×10^{15} for the reaction:

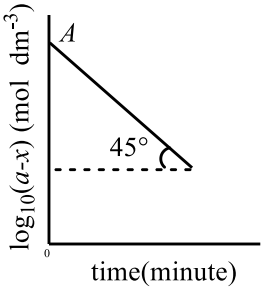
$$Cu_{(s)}+2Ag^{+}_{(aq)}\rightleftharpoons Cu^{2+}_{(aq)}+2Ag_{(s)}$$

The equilibrium constant for the reaction

$$\frac{1}{2}Cu^{2+}_{(aq)}+Ag_{(s)}\rightleftharpoons\frac{1}{2}Cu_{(s)}+Ag^{+}_{(aq)}$$

is $x\times10^{-8}$. The value of x is (Nearest integer)

48. For the first order reaction, variation of $\log_{10}(a-x)$ vs time (t) is shown below.



APEX JEE-MAIN 2025

For More Material Join: @JEEAdvanced_2025

Its half-life period is $\log p$. What is the value of p ?

49. 1 mole of ideal gas is allowed to expand reversibly and adiabatically from a temperature of $27°C$. The work done is $-3\ kJ\ mol^{-1}$. The final temperature of the gas is _____K (Nearest integer). Given $C_V = 20\ J\ mol^{-1}\ K^{-1}$.

50. How many molecules out of the following have non-zero value of dipole moments?
$CH_2Cl_2, CH_4, CCl_4, H_2O, CHCl_3$, p-dichlorobenzene, o-cresol, p-xylene,
$SCl_2, BF_3, IBr, HCHO$

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MATHS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. If P is orthogonal matrix of order 3x3 and $A=\begin{bmatrix}1 & 2 & 2\\ 2 & 1 & 2\\ 2 & 2 & 1\end{bmatrix}$ then the sum of positive divisors of $\left|P^TAP-2I\right|$ is equal to

A) 30 B) 40 C) 50 D) 60

52. The number of ways in which 16 identical toys distributed among 3 students such that each receives not less than 3 toys is

A) 9 B) 63 C) 36 D) 72

53. Let $A(1,-1,\lambda)$, $B(2,3,4)$ and $C(\lambda,5,2)$ be vertices of ΔABC if $\angle ABC=\frac{\pi}{2}$ then value of λ is

A) $\frac{2}{3}$ B) $\frac{3}{2}$ C) $\frac{3}{4}$ D) $\frac{4}{3}$

54. The mean and variance of a set of 15 numbers are 12 and 14 respectively. The mean and variance of another set of 15 numbers are 14 and σ^2 respectively. If the variance of all the 30 numbers in the two sets is 13, then σ^2 is equal to

A) 10 B) 11 C) 9 D) 12

55. The number of points of maximum and points of minimum of the function $f(x)=\max\{\sin x,\cos x\}$, $x\in(-2\pi,2\pi)$ are a and b respectively, then the value of $a+b$ is

A) 4 B) 9 C) 6 D) 7

APEX JEE-MAIN 2025

For More Material Join: @JEEAdvanced_2025

56. Let $Z=\dfrac{1+\cos\left(\frac{4\pi}{5}\right)-i\sin\left(\frac{4\pi}{5}\right)}{1+\cos\left(\frac{4\pi}{5}\right)+i\sin\left(\frac{4\pi}{5}\right)}$ and r,θ are respectively modulus and amplitude of Z^2 then $\dfrac{\theta}{r}$ equal to

A) $\dfrac{4\pi}{5}$ B) $\dfrac{2\pi}{5}$ C) $-\dfrac{4\pi}{5}$ D) $-\dfrac{2\pi}{5}$

57. The sum of positive roots of the equation $\dfrac{\sin^2 x+\cos^4 x}{\cos^2 x+\sin^4 x}=x^3-13x+13$ is

A) 8 B) 6 C) 5 D) 4

58. $ABCD$ is a quadrilateral with $\overrightarrow{AB}=\vec{a}$, $\overrightarrow{AD}=\vec{b}$ and $\overrightarrow{AC}=2\vec{a}+3\vec{b}$. If its area is α times the area of the parallelogram with $\overrightarrow{AB}$ and $\overrightarrow{AD}$ as adjacent sides, then α is equal to

A) 5 B) $\dfrac{5}{2}$ C) 1 D) $\dfrac{1}{2}$

59. If $1,\ \log_y x,\ \log_z y,\ -15\log_x z$ are in arithmetic progression then

A) $x=y^{-3}$ B) $x=z^3$ C) $y=z^{-2}$ D) $y=z^3$

60. If $\int\dfrac{\cos\theta}{3+\sin(2\theta)}d\theta=A\log\left(\dfrac{2+\sin\theta-\cos\theta}{2-\sin\theta+\cos\theta}\right)+B\tan^{-1}\left(\dfrac{\sin\theta+\cos\theta}{\sqrt{2}}\right)+C$ then B^2A is equal to

A) $\dfrac{1}{8}$ B) $\dfrac{1}{16}$ C) $\dfrac{1}{32}$ D) $\dfrac{1}{64}$

61. Let E be the ellipse $\dfrac{x^2}{9}+\dfrac{y^2}{4}=1$ and C be the circle $x^2+y^2=9$. Let P and Q be the points $(1,2)$ and $(2,1)$, respectively. Then:

A) Q lies inside C but outside E B) Q lies outside both C and E

C) P lies inside both C and E D) P lies inside C but outside E

APEX JEE-MAIN 2025

For More Material Join: @JEEAdvanced_2025

62. The solution of differential equation $y\cos\left(\frac{y}{x}\right)(xdy-ydx)+x\sin\left(\frac{y}{x}\right)(xdy+ydx)=0$ when $y(1)=\frac{\pi}{2}$ is $xy\sin\left(\frac{y}{x}\right)=P$ then P is equal to

A) $\frac{\pi}{3}$ B) $\frac{\pi}{4}$ C) $\frac{\pi}{6}$ D) $\frac{\pi}{2}$

63. In the binomial expansion of $\left(\sqrt{x}+\frac{1}{2.\sqrt[4]{x}}\right)^n$, the first three coefficients form an arithmetic progression, then sum of coefficients of all the terms is:

A) $\left(\frac{3}{2}\right)^5$ B) $\left(\frac{3}{2}\right)^6$ C) $\left(\frac{3}{2}\right)^7$ D) $\left(\frac{3}{2}\right)^8$

64. The distance of the point (1,2) from the line which is common to both lines $(x+2y-5)+\lambda(2x+y-7)=0$ and $(2x-3y-1)+\mu(x+y-3)=0$, $(\lambda,\mu\in R)$ is

A) 1 B) 2 C) 3 D) 4

65. The quadratic equation whose roots are $\cos\left(2\sin^{-1}\frac{2}{3}\right)$ and $\cos\left(2\tan^{-1}\frac{1}{3}\right)$ is $x^2+ax+b=0$ then $b-a$ is

A) $\frac{37}{45}$ B) $-\frac{37}{45}$ C) 1 D) -1

66. The function $f(x)=\frac{x}{\log_e x}$ is decreasing in

A) $(0,e)$ B) (e,∞) C) $(1,\infty)$ D) $(0,1)$

67. If $y=\frac{x^4+1}{x^2+\sqrt{2}x+1}$, at $x=\sqrt{2}$ the value of $\frac{\sqrt{2}y^{11}}{y^1}$ is equal to

A) 1 B) 2 C) -1 D) -2

68. Let R be the real line. Consider the following subsets of the plane $R\times R$:

S = {(x, y): y = x + 1 and 0 < x < 2}

T = {(x, y): x - y is an integer}. Which one of the following is true?

A) Both S and T are equivalence relations on R

APEX JEE-MAIN 2025

For More Material Join: @JEEAdvanced_2025

B) S is an equivalence relation on R but T is not

C) T is an equivalence relation on R but S is not

D) Neither S nor T is an equivalence relation of R

69. A card is drawn from a pack of 52 cards, the probability that the card drawn is a king or a heart card or a red card is

A) $\frac{6}{13}$ B) $\frac{7}{13}$ C) $\frac{8}{13}$ D) $\frac{29}{52}$

70. $\vec{r}$ is a vector of magnitude $\sqrt{51}$ and makes the same angle with $\vec{a}=\frac{\hat{i}-2\hat{j}+2\hat{k}}{3}$, $\vec{b}=\frac{-4\hat{i}-3\hat{k}}{5}$ and $\vec{c}=\hat{j}$, if $\vec{r}$ makes obtuse angle with x-axis then $\vec{r}.\left(\hat{i}+\hat{j}+\hat{k}\right)$ is equal to

A) 5 B) -5 C) 1 D) 3

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only.** **Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

71. The locus of midpoints of the chord of the circle $x^2-2x+y^2-2y+1=0$ which are of unit length is $(x-1)^2+(y-1)^2=k$, then 4k ----

72. For any solution (*a*, b, c) of the system of equations $x+y-2z+1=0$, $2x-y-z=1$ and $x-2y+z=2$ the equations $ax+by+c=2$ passes through fixed point (p,q) then the value of $p+q^2$ is ……..

73. Area of the region enclosed between the curves $x=y^2-1$ and $x=|y|\sqrt{1-y^2}$ is

74. The curve passing through (2,3) and satisfying the equation $\int_0^x ty(t)\,dt=x^2y(x),\ x>0$ is $xy=c$ then value of c is

APEX JEE-MAIN 2025

For More Material Join: @JEEAdvanced_2025

75. If the line $3x+2y=1$ touches the curve $y=f(x)$ at $x=\frac{1}{2}$ and

$$P=\lim_{x\to 0}\frac{x(x-1)}{f\left(\frac{e^{2x}}{2}\right)-f\left(\frac{e^{-2x}}{2}\right)}$$ then $50\sum_{r=1}^{\infty}P^r$

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